Name

Mathematics Advanced Year 11

Radians

Book 1

Version: 261021

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-04 applies trigonometry to solve problems involving geometric shapes

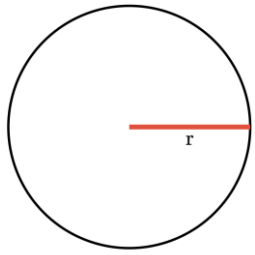
Radians

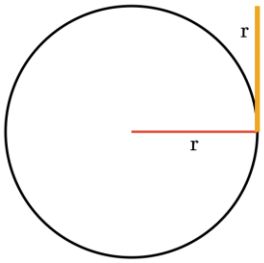
- Recognise that both ratios $\frac{\text{arc length}}{\text{circumference}}$ and $\frac{\text{area of sector}}{\text{area of circle}}$ are equal to $\frac{\theta}{\text{one revolution}}$ where θ is the angle at the centre of a circle subtended by the arc
- Define the angle size of θ in radian measure as the ratio $\frac{\text{arc length}}{\text{radius of a circle}}$
- Explain why $360° = 2\pi$ radians and 1 radian $= \frac{180}{\pi}$ degrees
- Convert between degrees and radians and find the exact sine, cosine and tangent ratios for integer multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$
- Graph $y = \sin x$, $y = \cos x$ and $y = \tan x$ over domains given in degrees or radians, showing intercepts with the x-axis and y-axis and any asymptotes, determine their domains and ranges, and period and amplitude where appropriate, and whether each is even or odd or neither
- Establish and use the formula $l = r\theta$ for the length l of arc subtending an angle θ in radians at the centre of a circle of radius r
- Prove and use the formula $A = \frac{1}{2}r^2\theta$ for the area of a sector with angle θ in radians at the centre of a circle of radius r
- Solve problems involving arc lengths and areas of major and minor sectors and segments

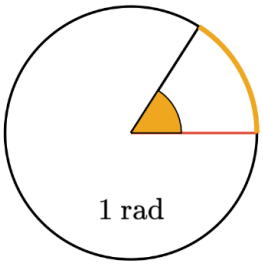
Radians

ⓘ Radians

- Ancient Babylonians invented degrees as an arbitrary unit.
  To connect degrees to their circular origin, we use **radians**.
- A **radian** is the **angle formed** when an **arc equals exactly 1 radius** in length.
- Radians represent a **dimensionless ratio** comparing **arc length** to **radius**.

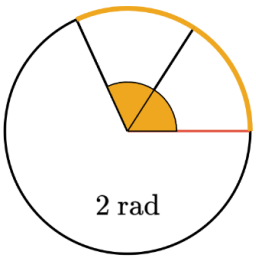


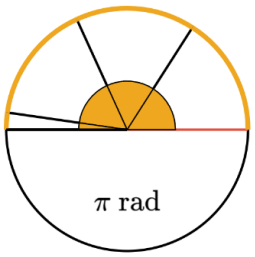


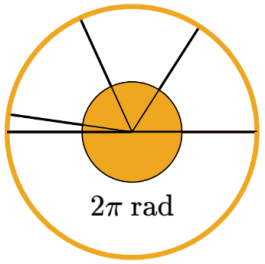


$180° = \pi$ radians

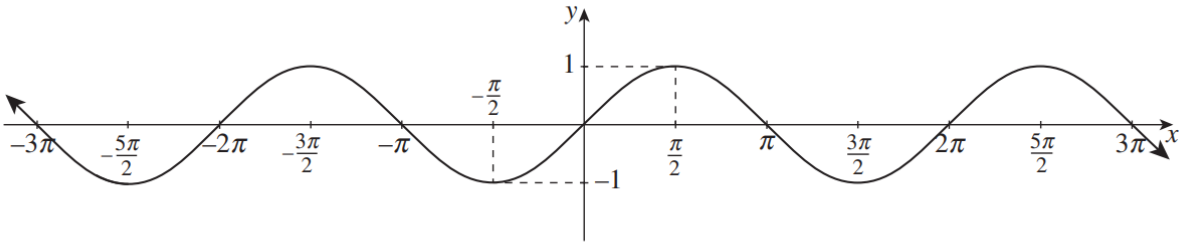
$360° = 2\pi$ radians



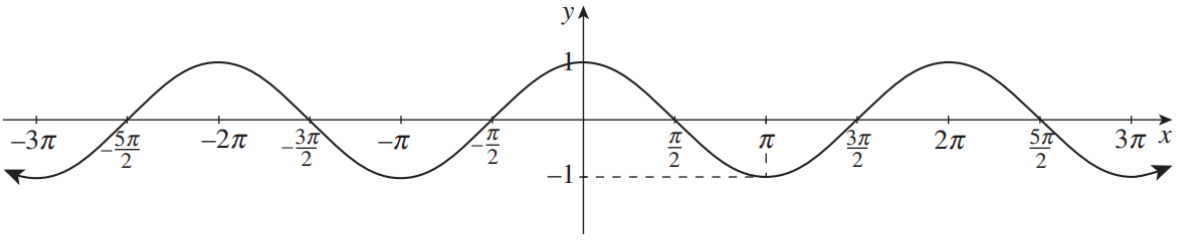




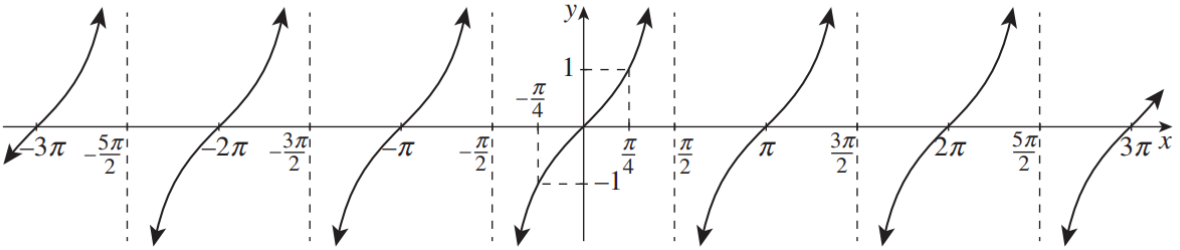
$y = \sin x$



$y = \cos x$



$y = \tan x$



Converting Radians and Degrees

- If an angle doesn't have a degree symbol, assume it is in radians.

π rad = 180°

radians to degrees:	degrees to radians:
$\times \frac{180°}{\pi}$	$\times \frac{\pi}{180°}$

Example Convert degrees to radians

60°

$60 \times \frac{\pi}{180}$

$= \frac{\pi}{3}$

495°

$495 \times \frac{\pi}{180}$

$= \frac{11\pi}{4}$

Guided Practice

Convert to radians.

a 30°

$\frac{\pi}{6}$

b 135°

$\frac{3\pi}{4}$

c 270°

$\frac{3\pi}{2}$

Example Convert radians to degrees

$\frac{\pi}{6}$

$\frac{\pi}{6} \times \frac{180}{\pi}$

$= 30°$

0.3

$0.3 \times \frac{180}{\pi}$

$\approx 17°$

Guided Practice

Convert to radians.

a $\frac{3\pi}{4}$

135°

b 20

1146

c 4π

720°

Key Ideas

1. π radians is ……°

1 Convert to degrees.

a	π	**b**	2π	**c**	3π	**d**	4π	**e**	5π
	180°		360°		540°		720°		900°
f	$\frac{\pi}{2}$	**g**	$\frac{3\pi}{2}$	**h**	$\frac{\pi}{3}$	**i**	$\frac{2\pi}{3}$	**j**	$\frac{4\pi}{3}$
	90°		270°		60°		120°		240°
k	$\frac{5\pi}{3}$	**l**	$\frac{\pi}{4}$	**m**	$\frac{3\pi}{4}$	**n**	$\frac{5\pi}{4}$	**o**	$\frac{7\pi}{4}$
	300°		45°		135°		225°		315°
p	$\frac{\pi}{6}$	**q**	$\frac{5\pi}{6}$	**r**	$\frac{7\pi}{6}$	**s**	$\frac{11\pi}{6}$		
	30°		150°		210°		330°		

2 Convert to radians in terms of π.

a	30°	**b**	45°	**c**	60°	**d**	90°	**e**	120°
	$\frac{\pi}{6}$		$\frac{\pi}{4}$		$\frac{\pi}{3}$		$\frac{\pi}{2}$		$\frac{2\pi}{3}$
f	135°	**g**	150°	**h**	180°	**i**	210°	**j**	225°
	$\frac{3\pi}{4}$		$\frac{5\pi}{6}$		π		$\frac{7\pi}{6}$		$\frac{5\pi}{4}$
k	240°	**l**	270°	**m**	300°	**n**	315°	**o**	330°
	$\frac{4\pi}{3}$		$\frac{3\pi}{2}$		$\frac{5\pi}{3}$		$\frac{7\pi}{4}$		$\frac{11\pi}{6}$

3 Convert to degrees.

a	$\frac{\pi}{2}$	**b**	$\frac{\pi}{3}$	**c**	$\frac{\pi}{4}$	**d**	$\frac{2\pi}{3}$	**e**	$\frac{5\pi}{3}$
	90°		60°		45°		120°		300°
f	$\frac{7\pi}{6}$	**g**	$\frac{4\pi}{3}$	**h**	$\frac{13\pi}{6}$	**i**	$\frac{3\pi}{2}$	**j**	$\frac{5\pi}{6}$
	210°		240°		390°		270°		150°
k	$\frac{11\pi}{6}$	**l**	$\frac{7\pi}{6}$	**m**	$\frac{2\pi}{5}$	**n**	$\frac{11\pi}{8}$	**o**	$\frac{17\pi}{10}$
	330°		210°		72°		247.5°		306°

4 Convert to radians in terms of π.

a	20°	**b**	22.5°	**c**	36°	**d**	100°	**e**	112.5°
	$\frac{\pi}{9}$		$\frac{\pi}{8}$		$\frac{\pi}{5}$		$\frac{5\pi}{9}$		$\frac{5\pi}{8}$

Exact Values in Radians

ⓘ Exact Values of the Trigonometric Ratios in Radians

	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{\pi}{6}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

⚠ You must memorise these exact values and be able to recall them quickly.

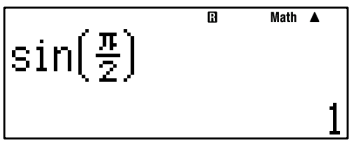
ⓘ Radians in a Calculator

- Your calculator is able calculate trigonometric functions that use radian measure.
- Always be aware of whether your calculator is in degrees mode or radians mode.
- If your answer is very small, or very close to 1, it is probably in the incorrect mode.

Casio FX-82 AU

1. SHIFT MODE
2. Select Rad

You should have a R on the top of the screen



Casio FX-8200 AU

1. Settings
2. Calc settings
3. Angle unit
4. Select Radian

You should have a R on the top of the screen

🔍 **Evaluate** Trigonometric functions in radians

Evaluate, using either your knowledge of exact values or a calculator.
All angle units are in radians. Round to 2 d.p. when it is not a known exact value.

a	$\cos\frac{\pi}{6}$	**b**	$\tan\frac{\pi}{3}$	**c**	$\sin\frac{\pi}{4}$
d	$\sin 4$	**e**	$\tan 1.3$	**f**	$\cos 0.5$

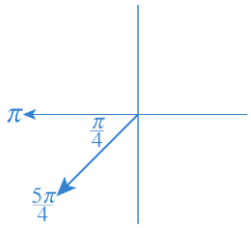
Angles of Any Magnitude using Radians

1. Determine the quadrant of the angle and sketch the angle.
2. Rewrite the function with the reference angle
3. Use ASTC to determine if the answer will be positive or negative.

Example Find exact value of a trigonometric function in radians

Find the exact value of these trigonometric functions.

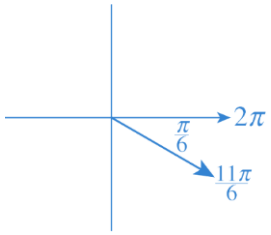
$\sin\frac{5\pi}{4}$

1. 

2. $\sin\frac{5\pi}{4} = -\sin\frac{\pi}{4}$

3. $= -\frac{1}{\sqrt{2}}$

$\cos\frac{11\pi}{6}$

1. 

2. $\cos\frac{11\pi}{6} = \cos\frac{\pi}{6}$

3. $= \frac{\sqrt{3}}{2}$

Guided Practice

Find the exact value of these trigonometric functions

a $\sin\frac{2\pi}{3}$

b $\tan\frac{7\pi}{6}$

c $\cos\left(-\frac{\pi}{6}\right)$

d $\tan\left(\frac{5\pi}{4}\right)$

1 Find the exact value of:

a $\sin\frac{\pi}{6}$ $\frac{1}{2}$	**b** $\sin\frac{\pi}{4}$ $\frac{1}{\sqrt{2}}$	**c** $\cos\frac{\pi}{6}$ $\frac{\sqrt{3}}{2}$	**d** $\tan\frac{\pi}{3}$ $\sqrt{3}$
e $\tan\frac{\pi}{4}$ 1	**f** $\cos\frac{\pi}{3}$ $\frac{1}{2}$	**g** $\tan\frac{\pi}{6}$ $\frac{1}{\sqrt{3}}$	**h** $\sin\frac{\pi}{3}$ $\frac{\sqrt{3}}{2}$
i $\tan\frac{\pi}{3}$ $\sqrt{3}$	**j** $\sin\frac{\pi}{3}$ $\frac{\sqrt{3}}{2}$	**k** $\sin\frac{\pi}{2}$ 1	**l** $\tan\frac{\pi}{2}$ undefined
m $1-\cos^2\frac{\pi}{4}$ $\frac{1}{2}$	**n** $\tan\frac{\pi}{3}\cos\frac{\pi}{3}$ $\frac{\sqrt{3}}{2}$	**o** $\sqrt{1+\tan^2\frac{\pi}{4}}$ $\sqrt{2}$	

2 With your calculator in radians mode, evaluate to 2 d.p. the following:

a $\sin 1$ 0.84	**b** $\cos 2$ -0.42	**c** $\tan 3$ -0.14
d $\sin 0.7$ 0.64	**e** $\cos 1.23$ 0.33	**f** $\tan 5.678$ -0.69

3 Using your knowledge of angles of any magnitude, find the exact value of:

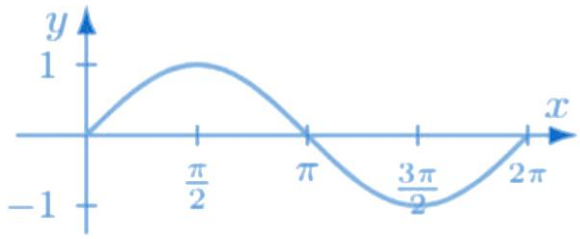
a $\sin\frac{2\pi}{3}$ $\frac{\sqrt{3}}{2}$	**b** $\cos\frac{2\pi}{3}$ $-\frac{1}{2}$	**c** $\cos\frac{5\pi}{6}$ $-\frac{\sqrt{3}}{2}$	**d** $\tan\frac{4\pi}{3}$ $\sqrt{3}$
e $\tan\frac{3\pi}{4}$ -1	**f** $\cos\frac{5\pi}{3}$ $\frac{1}{2}$	**g** $\sin\frac{5\pi}{4}$ $-\frac{1}{\sqrt{2}}$	**h** $\tan\frac{7\pi}{6}$ $\frac{1}{\sqrt{3}}$
i $\cos\frac{3\pi}{4}$ $-\frac{1}{\sqrt{2}}$	**j** $\sin\frac{5\pi}{6}$ $\frac{1}{2}$	**k** $\tan\frac{7\pi}{4}$ -1	**l** $\cos\frac{4\pi}{3}$ $-\frac{1}{2}$
m $\sin\frac{5\pi}{3}$ $-\frac{\sqrt{3}}{2}$	**n** $\cos\frac{13\pi}{6}$ $\frac{\sqrt{3}}{2}$	**o** $\tan\frac{11\pi}{3}$ $-\sqrt{3}$	**p** $\sin\frac{23\pi}{6}$ $-\frac{1}{2}$

4 Complete the table with exact values.

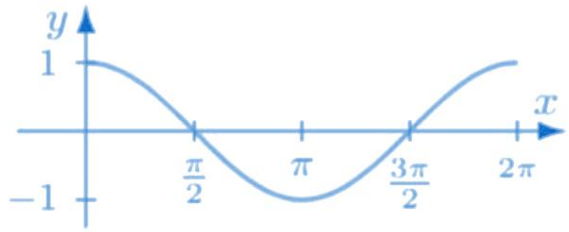
θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin\theta$	0	1	0	1	0
$\cos\theta$	1	0	1	0	1
$\tan\theta$	0	undefined	0	undefined	0

The 3 graphs below show the sketches of $y = \sin x$, $y = \cos x$, $y = \tan x$ in domain $x \in [0, 2\pi]$. Identify the graphs then label all key relevant points using your tables .

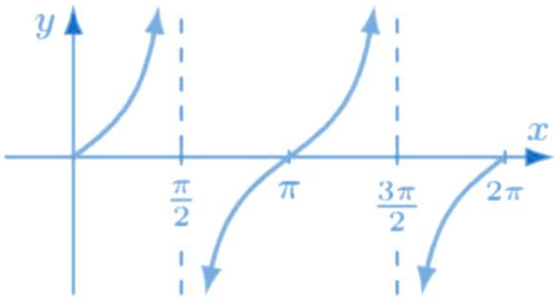
$y = \sin x$



$y = \cos x$

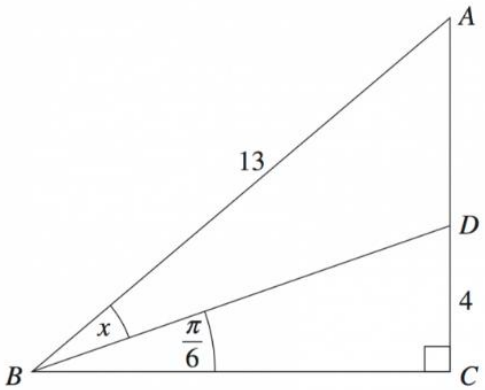


$y = \tan x$



5 **2013 HSC Advanced Band 5**

Using the sine rule, or otherwise, find the exact value of $\sin x$.



Approach: Use sine rule in triangle ΔADB by finding $\angle ADB$ and AD first.

$\angle BDC = \pi - \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$

$\therefore \angle ADB = \frac{2\pi}{3}$

$\tan\frac{\pi}{6} = \frac{4}{BC}; \quad BC = 4\sqrt{3}$

Using Pythagoras in ΔABC,

$AC = \sqrt{13^2 - (4\sqrt{3})^2} = 11$

$\therefore AD = AC - DC = 7$

Using sine rule:

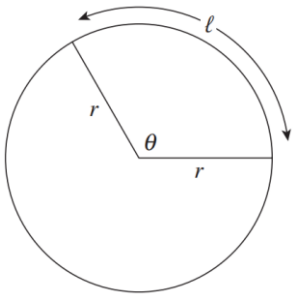
$\frac{\sin x}{7} = \frac{\sin\frac{2\pi}{3}}{13}$

$\sin x = \frac{7\sin\frac{2\pi}{3}}{13} = \frac{6\sqrt{3}}{26}$

Arc Length and Sector Area using Radians

ⓘ Arc Length Formula

- The arc length formula using radians is much “neater”.
  This neatness is essential in the future in trigonometric calculus.

Arc length using degrees	Arc length using radians	
$l = \frac{\theta}{360°} \times 2\pi r$	$l = r\theta$	

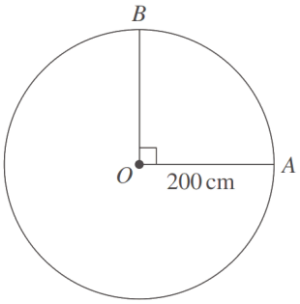
Example Calculate arc length using radians

What is the length of an arc subtending a right angle at the centre of a circle of radius 200 cm?

$l = r\theta$

$= 200 \times \frac{\pi}{2}$

$= 100\pi$ cm

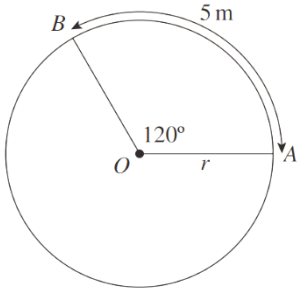


What is the radius of a circle in which an arc of length 5m subtends an angle of 120° at the centre?

$l = r\theta$

$5 = r \times \frac{2\pi}{3}$

$\frac{15}{2\pi} = r$



Guided Practice

a Find the arc length subtending an angle of $\frac{\pi}{6}$ at the centre of circle of radius 18 cm

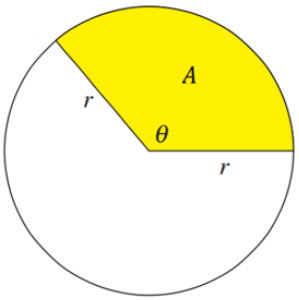
3π cm

b A circle has radius 6 cm. Find the length of an arc of this circle that subtends an angle at the centre of 2 radians.

12 cm

ⓘ Sector Area Formula

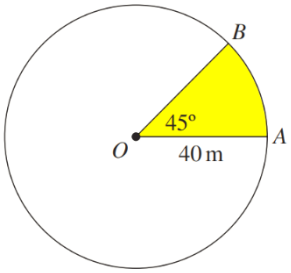
- The arc length formula using radians is much “neater”.
  This neatness is essential in the future in trigonometric calculus.

Sector area using degrees	Sector area using radians	
$A = \frac{\theta}{360°} \times \pi r^2$	$A = \frac{1}{2} r^2 \theta$	

Example Calculate sector area using radians

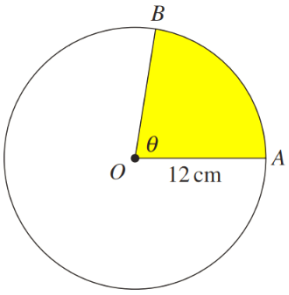
What is the area of a sector subtending an angle of 45° at the centre of a circle of radius 40m?

$$A = \frac{1}{2} r^2 \theta$$
$$= \frac{1}{2} \times 40^2 \times \frac{\pi}{4}$$
$$= 200\pi \text{ m}^2$$



A circular cake has radius 12 cm. What is the angle subtended at the centre by a sector of area 100 cm^2? Answer to the nearest degree.

$$A = \frac{1}{2} r^2 \theta$$
$$100 = \frac{1}{2} \times 12^2 \times \theta$$
$$\frac{200}{12^2} = \theta$$



$80° \approx \theta$

Guided Practice

a What is the area of a sector subtending an angle of $\frac{\pi}{4}$ at the centre of a circle of radius 4 cm?

2π cm^2

b A circle has radius 8 cm. Find the area of a sector of this circle that subtends an angle at the centre of 3 radians.

96 cm^2

1 A sector has radius 5 cm. If the angle subtended at the centre is 2 radians, find the:

a Arc length

10 cm

b Area of the sector

25 cm^2

2 A sector has radius 6 cm. If the angle subtended at the centre is 120°, find in exact form:

a Arc length

4π cm

b Area of the sector

12π cm^2

3 A sector has radius 3 cm. If the angle subtended at the centre is 150°, find in exact form:

a Arc length

$\frac{5\pi}{2}$ cm

b Area of the sector

$\frac{15\pi}{4}$ cm^2

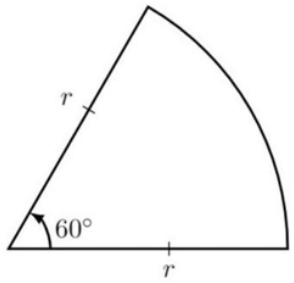
4 A sector has radius 10 cm. If the arc length is 8cm, find in exact form the angle subtended at the centre in radians.

$\frac{4}{5}$

5 A sector has radius 5 cm and area 10π cm^2. Find the arc length of the sector in exact form.

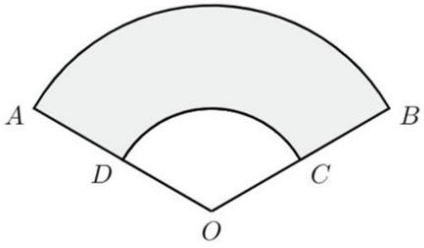
4π cm

6 The sector has an area of 15π cm^2. What is the value of r?



$3\sqrt{10}$ cm

7 A car windscreen wiper traces out an area ABCD where AB and CD are arcs of circles with centre O and radii 30 cm and 15 cm respectively. $\angle AOB = 120°$.
Find the area of ABCD in exact form.

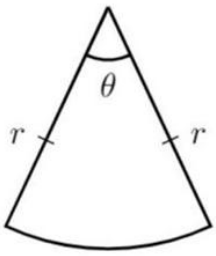


225π cm^2

8 Let the sector subtended by angle θ have area 625π cm^2.

a Find the value of r if $\theta = \frac{\pi}{5}$.

b Find the value of θ if $r = 8$ cm.

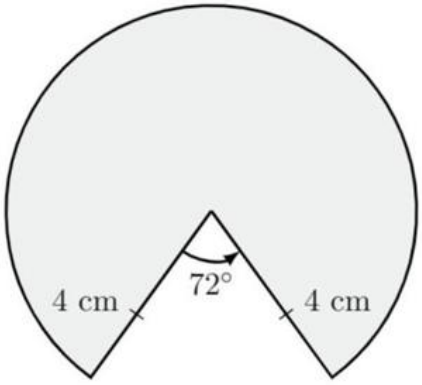


a) $25\sqrt{10}$ cm b) $\frac{625}{32}\pi$

9 A sector subtending $72°$ is cut from a circle of radius 4 cm. Find the

a shaded area.

b length of remaining arc.

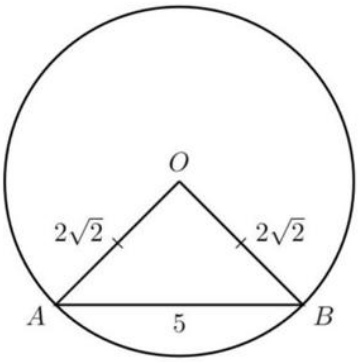


a) $\frac{64}{5}\pi$ cm^2 b) $\frac{32}{5}\pi$ cm

10 A chord of length 5 cm is drawn in a circle with centre O and radius $2\sqrt{2}$ cm.

a Use cosine rule to find $\angle AOB$ is radians, correct to 3 d.p.

b Hence find the arc length cut off by the chord, correct to 2 d.p.



a) 2.168 b) 6.13 cm

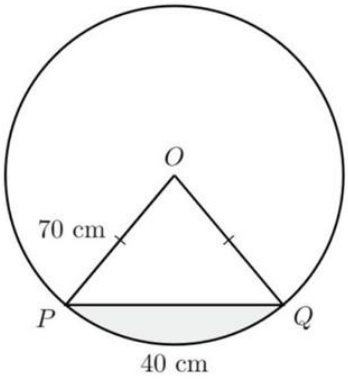
11 The diagram below shows an arc PQ of length 40 cm, subtended by some angle in a circle with radius 70 cm. Find:

a $\angle POQ$ in radians.

b Area of sector POQ.

c Area of ΔPOQ to the nearest cm^2

d Shaded area correct to nearest cm^2



a) $\frac{4}{7}$

b) 1400 cm^2

c) 1325 cm^2

d) 75 cm^2

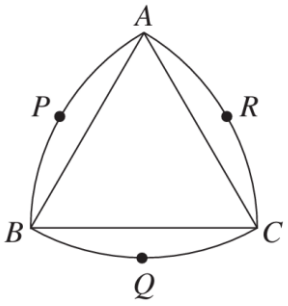
12 Triangle ABC is equilateral with side length 2 cm. Circular arcs AB, BC and CA have centres C, A and B respectively. Answer each part in exact form.

a Find the length of the arc AB.

b Find the area of the sector CAPBC.

c Find the length of the perimeter APBQCRA.

d Find the area of △ABC and hence find the area enclosed by the perimeter APBQCRA.



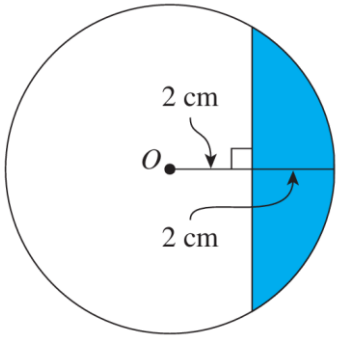
a) $\frac{2\pi}{3}$ cm

b) $\frac{2\pi}{3}$ cm^2

c) 2π cm

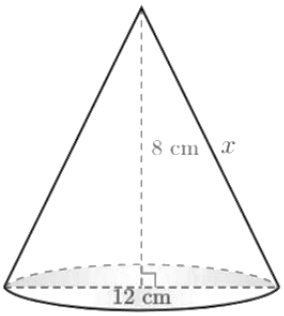
d) $\sqrt{3}$ cm^2, $2(\pi - \sqrt{3})$ cm^2

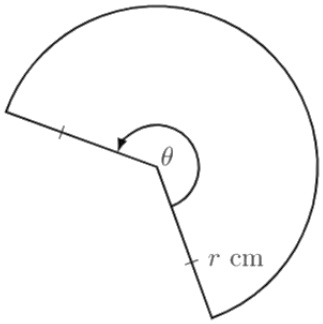
13 Find the exact area of the shaded region of the circle shown in the diagram.



$\frac{4}{3}(4\pi - 3\sqrt{3})$ cm²

14 A cone is made from a sector with radius r and angle θ by joining the two straight-line edges together.





a Find the radius of the cone.

b Hence find the value of r. hint: what does r become on the cone?

c Find the arc length of the sector. hint: what does the arc length become on the cone?

d Find the sector angle θ in radians.

a) 6 cm

b) 10 cm

c) 12π cm

d) $\frac{6\pi}{5}$

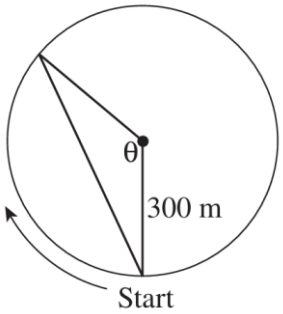
15 An athlete runs at a steady 4 m/s around a circular track of radius 300 metres. She runs clockwise, starting at the southernmost point.

a What distance around the track has she run after 3 minutes?

b What angle does this distance subtend at the centre?

c How far is she from her start, in a direct line across the field, correct to the nearest 0.01 metre?

d What is her true bearing from the centre then, correct to the nearest minute?



a) 720 metres

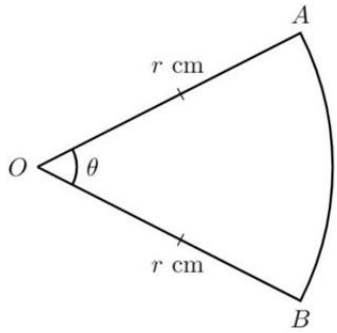
b) 2.4 radians

c) 559.22 metres

d) 317° 31' T

16 The sector OAB has an area of $\frac{\pi}{2}$ cm^2. The arc length is $\frac{\pi}{4}$ cm.

By solving simultaneously, find the exact values of r and θ



$r = 4$ cm, $\theta = \frac{\pi}{16}$

17 The area of a sector is $\frac{3\pi}{10}$ cm^2 and the arc length cut off by the sector is $\frac{\pi}{5}$ cm.

Find the exact value of r and θ.

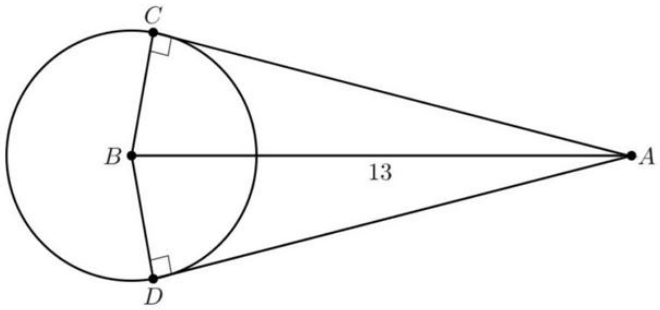
$r = 3$ cm, $\theta = \frac{\pi}{15}$

18 AC and AD are equal tangents to the circle of radius 5 cm.
The point A is 13 cm from the centre B.
BC is perpendicular to AC and BD is perpendicular to AD.

a Find $\angle ABC$ in radians, to 3 d.p.

b Find minor arc CD, to 2 d.p.

c Find major arc CD, to 2 d.p.



a) 1.176

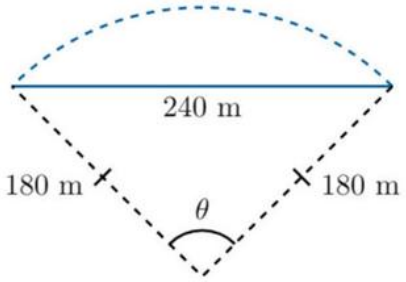
b) 11.76

c) 19.66

19 A straight road was constructed to cut a dangerous bend on a country road. It was found that the bend was part of an arc of radius 180 m and the straight road was 240 m long.

a Use cosine rule to find θ in radians, to 2 d.p.

b Find the distance by which the old road was shortened, to the nearest metre.



a) 1.46

b) 23 m

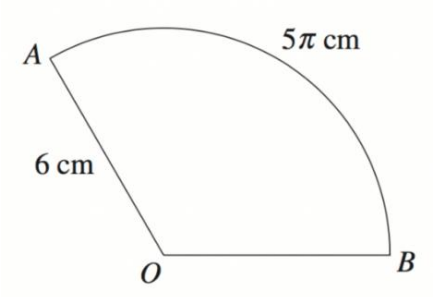
20 A bicycle wheel of radius 36 cm turns through an angle of $\frac{2\pi}{3}$ radians in 0.16 seconds. At what speed is the bicycle being ridden, in metres per minute? Answer in exact form.

Speed = distance ÷ time

$$= \frac{0.36 \times \frac{2\pi}{3} \text{ metres}}{\left(\frac{0.16}{60}\right) \text{ minutes}} = 90\pi \text{ metres/min}$$

21 **2004 HSC Advanced Band 4**

Calculate the exact area of sector AOB



$l = r\theta$

$5\pi = 6\theta$

$\theta = \frac{5\pi}{6}$

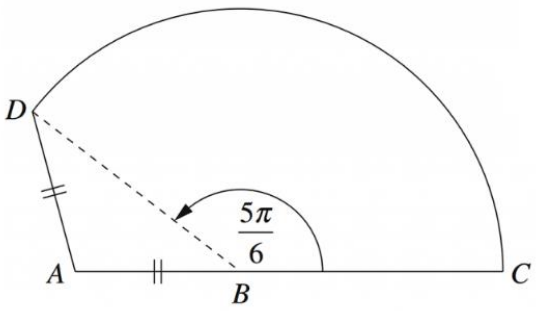
$A = \frac{1}{2}r^2\theta$

$= \frac{1}{2}(6)^2\left(\frac{5\pi}{6}\right)$

$= 15\pi \text{ cm}^2$

22 2006 HSC Advanced Band 4

ABCD represents a garden. Points ABC lie on a straight line and $AB = AD = 3$ m.



a Show that $\angle DAB = \frac{2\pi}{3}$

$\angle DBA = \pi - \frac{5\pi}{6} = \frac{\pi}{6}$ (angles on a straight line)

$\angle BDA = \angle DBA = \frac{\pi}{6}$ (angles opposite equal sides of isosceles triangle)

$\angle DAB = \pi - \frac{\pi}{6} - \frac{\pi}{6}$ (angle sum of a triangle)

$= \frac{2\pi}{3}$

b Find the length of BD.

Using cosine rule

$BD = \sqrt{3^2 + 3^2 - 2(3)(3)\cos\frac{2\pi}{3}}$

$= \sqrt{27}$

$= 3\sqrt{3}$ m

c Hence find the area of the garden.

Area $\Delta ABD = \frac{1}{2}(3)(3)\sin\frac{2\pi}{3} = \frac{9\sqrt{3}}{4}$

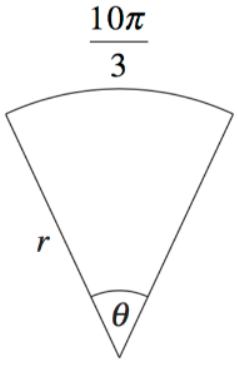
Area of sector $= \frac{1}{2}r^2\theta = \frac{1}{2}(\sqrt{27})^2\left(\frac{5\pi}{6}\right)$

$= \frac{45}{4}\pi$ m^2

Area of garden $ABCD = \frac{9\sqrt{3}}{4} + \frac{45\pi}{4} = \frac{9\sqrt{3}+45\pi}{4}$ m^2

23 2008 HSC Advanced Band 5

The diagram shows a sector with radius r and angle θ where $0 \leq \theta \leq 2\pi$. The arc length is $\frac{10\pi}{3}$.



a Show that $r \geq \frac{5}{3}$

$r\theta = \frac{10\pi}{3}$

$\theta = \frac{10\pi}{3r}$

As $0 \leq \theta \leq 2\pi$

$0 \leq \frac{10\pi}{3r} \leq 2\pi$

$\frac{10\pi}{3} \leq 2\pi r$

$\frac{5}{3} \leq r$

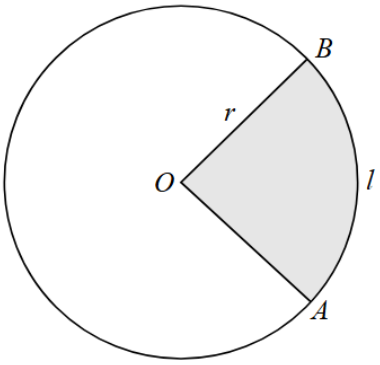
As required

b Calculate the area of the sector when $r = 4$

$A = \frac{20\pi}{3}$ units^2

24 HSC Sample Question Advanced Band 5

The ratio of the radius r to the arc-length l is 2:5. Find the area of the sector in terms of r.



By definition, radian is the ratio of arc-length to radius.

$\therefore \theta = \frac{l}{r} = \frac{5}{2}$

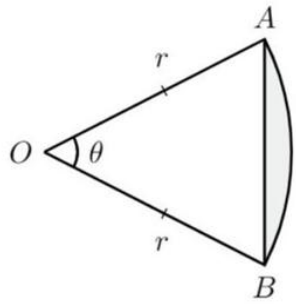
$A = \frac{1}{2}r^2\left(\frac{5}{2}\right)$

$A = \frac{5r^2}{4}$

25

The shaded region is called a segment. Prove that its area is given by the formula

$$A = \frac{1}{2}r^2(\theta - \sin\theta)$$



Area of triangle: $\frac{1}{2}r^2\sin\theta$

Area of sector: $\frac{1}{2}r^2\theta$

Area of segment = Sector – Triangle

$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2(\sin\theta)$

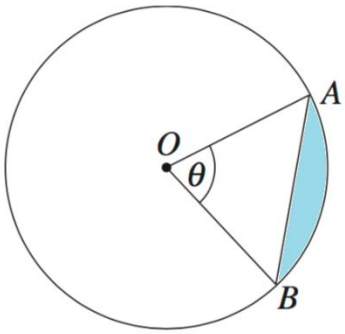
$= \frac{1}{2}r^2(\theta - \sin\theta)$

26 2009 HSC Advanced Band 5

The triangle ΔAOB is inscribed inside a circle of radius 2 cm. The area of the triangle is $\sqrt{3}$ cm^2.

a Find two possible values of θ.

b Suppose that $\theta = \frac{\pi}{3}$, find the exact perimeter of the minor segment OAB.



a) using $A = \frac{1}{2}ab\sin C$

$\sqrt{3} = \frac{1}{2}(2)(2)\sin\theta$

$\therefore \theta = \frac{\pi}{3}$ or $\frac{2\pi}{3}$

b) using cosine rule, straight line $AB = 2$

arc length $AB = \frac{2\pi}{3}$ cm

$\therefore P = 2 + \frac{2\pi}{3}$ cm